

ON COFACTUALS

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WHAT IS THE ECONOMIC QUESTION?

How can marginal policy-world distributions discipline the distribution of policy gains?

- A gain compares two potential outcomes carried by the same unit.
- Marginal world distributions leave the cross-world pairing open.
- The paper asks when a shared-component structure turns those marginals into a policy-gain distribution.

WHAT DOES A TWO-POLICY EXAMPLE LOOK LIKE?

- With one policy label, $Y_{\emptyset} = C_{\emptyset}$ and $Y_{\{1\}} = C_{\emptyset} + C_{\{1\}}$.
- The unit-level policy gain is $Y_{\{1\}} - Y_{\emptyset} = C_{\{1\}}$.
- With several labels, higher-order components record interactions that appear only when their full label set is active.

WHICH COORDINATES CARRY CROSS-WORLD DEPENDENCE?

- Worlds are indexed by policy sets $S \subseteq \mathcal{P}$.
- Möbius inversion defines components $C_A = \sum_{B \subseteq A} (-1)^{|A|-|B|} \gamma_B$.
- Cofactual factorization requires the family $\{C_A\}$ to be mutually independent.

WHY CAN IDENTICAL MARGINALS IMPLY DIFFERENT GAINS?

World marginals



Cross-world coupling



Gain distribution

WHEN DO THE MARGINALS ADMIT A FACTORIZED COUPLING?

Theorem (Marginal-to-joint characterization)

Under local exponential integrability, a factorized coupling exists if and only if every Möbius-transformed cumulant generating function H_A is an admissible CGF. Within the class, the component laws and joint law are unique.

The characterization converts a family of marginal laws into a check on component CGFs.

HOW ARE THE COMPONENT LAWS RECOVERED?

$$H_A(t) = \sum_{B \subseteq A} (-1)^{|A|-|B|} K_B(t)$$

- K_B is the cumulant generating function of world B .
- H_A is the candidate cumulant generating function of component C_A .
- Admissibility of every H_A supplies the existence check and the component laws.

HOW DOES THE CHARACTERIZATION CONSTRUCT THE JOINT LAW?

- Treat each admissible transformed CGF as the law of one independent component.
- Assemble each world by $Y_S = \sum_{A \subseteq S} C_A$.
- The construction matches the supplied marginals and fixes the joint law within the class.

WHAT BECOMES EMPIRICALLY DECIDABLE?

- Marginal counterfactual laws imply a candidate family of interaction cumulants through Möbius inversion.
- Admissibility determines whether those candidates form probability laws, and uniqueness fixes the factorized joint law within the class.
- The same construction supplies specification checks and identifies the additional intersections needed for a declared target.

WHAT BECOMES IDENTIFIED INSIDE THE CLASS?

Proposition (Marginal-to-joint identification)

Admissible transformed CGFs recover every component law, every cross-world joint cumulant, and the joint law within the factorized class. They therefore recover each policy-gain law $Y_S - Y_T$.

The identifying content is class relative and carries the full pairing required by gain distributions.

WHICH RESTRICTIONS CAN THE OBSERVATION SCHEME ASSESS?

- Marginal samples assess admissibility and variance-surface inequalities.
- Paired world draws assess intersection covariance and higher-order cumulant restrictions.
- State-adjusted sampling assesses conditional covariance restrictions under the declared controls.

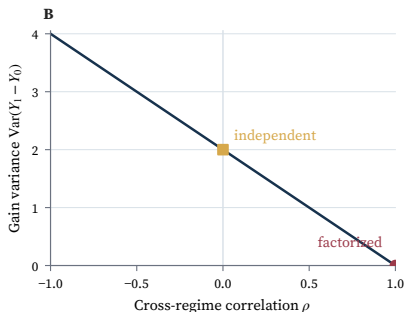
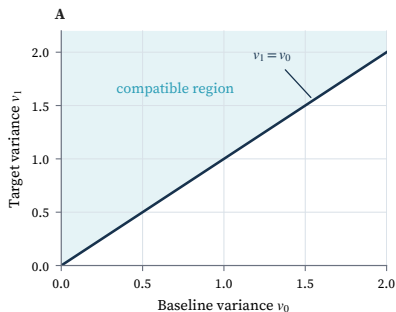
HOW DO POPULATION RESTRICTIONS BECOME SAMPLE OBJECTS?

- Variance inequalities use marginal samples because each world variance is observed separately.
- Intersection covariances use paired worlds because the same unit must be observed across regimes.
- Conditional versions add state adjustment before the covariance restriction is formed.

WHERE DOES THE SHARED-COMPONENT STRUCTURE COME FROM?

- A modular Bellman model propagates independent reward components through a fixed policy.
- The same result covers optimal values when the fixed policy is optimal in every world almost surely.
- Reoptimization examples mark the boundary where policy changes alter the component map.

WHICH CONCLUSIONS USE MARGINALS, AND WHICH USE PAIRED WORLDS?



Main-text identification map. Marginal compatibility, class-relative coupling recovery, and paired restrictions remain separate.

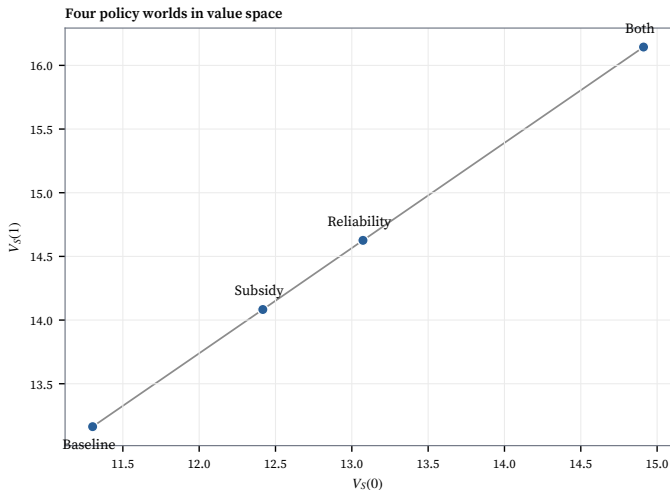
HOW DOES IMPERFECT ORTHOGONALITY CHANGE GAIN DISPERSION?

Proposition (Variance-error bound)

For a linear policy contrast, the error from imposing component orthogonality is bounded by the squared contrast loading times the operator norm of the component-covariance defect.

The bound links a measurable departure from the model to the dispersion of policy gains.

HOW DOES THE DYNAMIC EXAMPLE TRANSLATE COMPONENTS INTO VALUE?



Appendix value geometry for the Bellman illustration.

WHAT DOES THE FINITE-SAMPLE EXERCISE EVALUATE?

- Wald statistics target selected paired covariance restrictions.
- The simulations examine size and power under the declared paired sampling design.
- Calibration evidence supports the testing procedure; it does not validate the full factorized class from marginals alone.

WHICH LITERATURES HELP PLACE THE QUESTION?

- Chernozhukov, Fern'andez-Val, and Melly (2013): counterfactual distributions from structural distributional inputs.
- Heckman and Vytlacil (2005): policy-relevant treatment objects organized through a common structure.
- Christensen and Connault (2023): counterfactual sensitivity to maintained and perturbed distributional assumptions.
- This paper isolates the cross-world pairing delivered by independent Möbius components.

HOW FAR DOES THE RESULT TRAVEL?

- The full CGF result uses local exponential integrability.
- Joint-law recovery is unique within the factorized class.
- Moment and sampling results address the restrictions observed under each acquisition design.

WHAT SHOULD THE READER CARRY FORWARD?

1. Policy gains require a pairing across worlds.
2. Möbius-transformed CGFs characterize a shared-component pairing class.
3. The observation scheme determines which implications can be assessed.

TECHNICAL APPENDIX

WHICH DEFINITIONS AND ASSUMPTIONS ORGANIZE THE PAPER?

- Assumption: Local exponential integrability
- Definition: Counterfactual system
- Definition: Cofactual component
- Definition: Cofactual factorization
- Definition: Admissible CGF

HOW DO THE MAIN RESULTS BUILD ON ONE ANOTHER?

Marginal-to-joint characterization



Marginal-to-joint identification



Variance-error bound

WHICH FORMAL RESULTS SUPPORT THE ARGUMENT?

- Theorem: Marginal-to-joint characterization
- Theorem: Intersection-cumulant characterization
- Theorem: Cofactual Wald test
- Proposition: Cofactual representation
- Proposition: Modular Bellman propagation
- Proposition: Meet-kernel factorization
- Proposition: Variance-surface characterization

WHICH SCOPE CONDITIONS GOVERN THE CLAIMS?

- The full CGF result uses local exponential integrability.
- Joint-law recovery is unique within the factorized class.
- Moment and sampling results address the restrictions observed under each acquisition design.